

ISC COMPUTER SCIENCE – TEST II (60 Marks)

Syllabus Weightage Based on Class 11 questions asked in Class 12 Foundations
(Logic & Hardware, Boolean Axioms, Recursion, Arrays, Strings, Packages, Basic Input/Output)

SECTION A – MCQs (20 × 1 = 20 marks)

1. The dual of the Boolean expression $(A + 0 = A)$ is:
a) $(A \cdot 1 = A)$ b) $(A + 1 = 1)$ c) $(A \cdot 0 = 0)$ d) $(A \cdot 1 = 1)$
2. Which law is represented by $(A + A = A)$?
a) Idempotent Law b) Associative Law c) Commutative Law d) Absorption Law
3. The contrapositive of “If P then Q” is:
a) If Q then P b) If not Q then not P c) If not P then Q d) Q implies P
4. NAND is called a universal gate because:
a) It has three inputs
b) Any circuit can be built using only NAND gates
c) It gives inverted output
d) None of the above
5. Output of XOR gate for inputs (1, 0):
a) 0 b) 1 c) Undefined d) Same as AND
6. In recursion, the program uses which structure internally?
a) Queue b) Stack c) Heap d) Array
7. A 2D array stored in **row-major order** places elements in memory:
a) Column by column b) Row by row c) Diagonally d) Arbitrarily
8. Which String method returns an integer based on lexicographic comparison?
a) equals() b) compareTo() c) length() d) valueOf()
9. A Java package helps in:
a) Speeding execution
b) Avoiding naming conflicts
c) Adding memory
d) Storing binary data
10. The NOT gate gives output 0 when input is:
a) 1 b) 0 c) Both d) None
11. Which of these best defines a tautology?
a) Always false b) Sometimes true
c) Always true d) Depends on input
12. Which method will convert a string to uppercase?
a) toUpperCase() b) toCaps() c) upper() d) setUpperCase()
13. Address calculation requires:
a) Base address and width b) Only width
c) Only row number d) Only column number
14. Which of the following is immutable in Java?
a) String b) StringBuffer c) ArrayList d) int[]

15. What is the binary equivalent of decimal 9?
a) 1000 b) 1001 c) 1110 d) 1010
16. In propositional logic, implication $(P \rightarrow Q)$ is equivalent to:
a) $(P + Q)$ b) $(\bar{P} + Q)$ c) (PQ) d) $(P + \bar{Q})$
17. The base case in recursion is used to:
a) Increase call depth
b) Terminate recursive calls
c) Print output faster
d) Convert iteration
18. `import java.util.*; imports:`
a) All Java packages
b) All classes in util
c) Only Scanner
d) Only Math class
19. In a binary file, data is stored as:
a) Characters b) Unicode blocks
c) Raw bytes d) ASCII only
20. Which of the following is NOT a basic gate?
a) AND b) NOR c) OR d) NOT
-

SECTION B – Assertion–Reasoning ($5 \times 2 = 10$ marks)

(Use: a) Both true, R explains A
b) Both true, R doesn't explain A
c) A true, R false
d) A false, R true)

21. **A:** De Morgan's law helps convert expressions into NAND/NOR realizations.
R: De Morgan's law transforms AND into OR with complemented variables.
22. **A:** A recursive function without a proper base case may cause stack overflow.
R: Every recursive call consumes a new activation record on the call stack.
23. **A:** Strings in Java are immutable.
R: Changing a String creates a new String object in memory.
24. **A:** In row-major storage, all elements of a row are stored contiguously.
R: Row-major storage places elements column by column.
25. **A:** The NAND gate can implement NOT, AND and OR operations.
R: It combines AND logic followed by a NOT operation.
-

SECTION C – One Sentence Answers ($5 \times 2 = 10$ marks)

26. Define the Principle of Duality in Boolean Algebra.

- 27. What is a contrapositive of a proposition?
 - 28. Why is the String class considered immutable?
 - 29. What is meant by base address of an array?
 - 30. What is the purpose of the recursive base case?
-

SECTION D – Programming / Analytical Questions (4 × 5 = 20 marks)

(No file-handling programs as requested.)

31. Recursion – Output Tracing

Consider the function:

```
int fun(int n) {  
    if(n <= 1)  
        return 1;  
    return n * fun(n - 2);  
}
```

Find the output for fun(7) with proper stack expansion.

32. Recursion – Algorithmic Design

Write a recursive method to check whether a given number is **Evil** (even number of 1s in binary).

(Hint based on digit/binary logic emphasized in the uploaded document .)

33. Arrays – Address Calculation Problem

A matrix M[5][6] is stored in **column-major order**.

Each element requires **2 bytes**.

Base address = **1000**.

Find the address of M[3][2].

34. String Manipulation – Analytical Reasoning

Predict the output:

```
String s = "ComputeR";  
String t = s.substring(3).replace('e', 'E');  
System.out.println(t.compareTo("uteR"));
```

Explain each step briefly.

35. Packages – Conceptual Application

Explain with example syntax:

- how to declare a user-defined package
- how another class can import and access it.